

ITC101- AI PRINCIPLES AND TECHNIQUES

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CLASS TASK

1. What is Utf 8
2. Pros and cons of CSV and xlsx
3. Learn about Encoding
4. Create an account on 'Hugging face'

1)What is Utf 8

UTF-8(*bit Unicode transformation Format) is a variable length character encoding system that translates any unicode character into a unique sequence of 1 to 4 bytes.

Computers don't understand letters, words, or emojis—they only understand numbers, specifically 0s and 1s. **UTF-8** is essentially a universal translation system that maps every character in human language to a specific code of 0s and 1s so computers can process and display it properly.

Think of Unicode as a massive digital dictionary that assigns a unique number to every character across the globe—from the plain English letter "A" to complex Japanese characters, mathematical symbols, and emojis. UTF-8 is the smart encoding system that actually turns those dictionary numbers into binary computer bytes.

What makes UTF-8 brilliant is that it uses a **variable length** (1 to 4 bytes) depending on the character:

- Simple English characters (like standard ASCII) only take **1 byte**, keeping file sizes tiny.
- More complex international characters take **2 or 3 bytes**.
- Emojis and rare historical scripts take **4 bytes**.

Code Point Range	Bytes	Byte 1	Byte 2	Byte 3	Byte 4	Examples
U+0000 to U+007F	1	0xxxxxxx	—	—	—	A, b, 5, \$
U+0080 to U+07FF	2	110xxxxx	10xxxxxx	—	—	é, ñ, Ω, μ
U+0800 to U+FFFF	3	1110xxxx	10xxxxxx	10xxxxxx	—	€, あ, 文, ₹
U+10000 to U+10FFFF	4	11110xxx	10xxxxxx	10xxxxxx	10xxxxxx	🤪, 🚀, 🔥, 🌹

Small Example: Encoding the character A vs. €

- **The Letter A (1 Byte)**
 - Unicode Code Point: U+0041
 - Fits into the **1-byte** range (0xxxxxxx).
 - Binary stored on disk: 01000001 (Only takes 8 bits / 1 byte).
- **The Euro Sign € (3 Bytes)**
 - Unicode Code Point: U+20AC
 - Fits into the **3-byte** range (1110xxxx 10xxxxxx 10xxxxxx).
 - Binary stored on disk: 11100010 10000010 10101100 (Takes 24 bits / 3 bytes).

Before UTF-8, opening a foreign document or webpage often resulted in broken, gibberish characters because systems used different regional rules. UTF-8 solved this by becoming the single universal standard, making sure text looks identical across every device, operating system, and web browser worldwide.

2. Pros and cons of CSV and xlsx

CSV (Comma-Separated Values) is a lightweight, plain-text file format where data rows are written line-by-line and fields are separated by commas. It serves as a standard format for raw data storage, database transfers, and backend processing.

- **Pros:**
 - **Universal Compatibility:** Opens easily in any text editor, programming language, database, or spreadsheet program.
 - **Small File Size:** Contains zero layout metadata, keeping files lightweight and fast to read or write.
 - **Developer-Friendly:** Plain-text structure integrates seamlessly with scripts and allows line-by-line Git version tracking.
 - **No Software Lock-In:** Free to generate and manipulate without needing proprietary software or special libraries.
- **Cons:**
 - **No Formatting or Visuals:** Cannot store cell styling, colors, custom fonts, or embedded charts.
 - **No Formulas or Functions:** Flattens dynamic calculations down to static text values upon saving.
 - **Single Sheet Limit:** Restricted to storing only one table per file; cannot support multiple workbook tabs.

- **Data Alteration Risks:** Spreadsheet apps can auto-strip leading zeros or convert numbers to scientific notation.

XLSX (Microsoft Excel Open XML Spreadsheet) is a feature-rich, XML-based spreadsheet format designed by Microsoft for interactive data analysis, reporting, and visualization.

- **Pros:**

- **Advanced Functionality:** Supports dynamic formulas, built-in functions, pivot tables, and macro automation.
- **Rich Presentation:** Allows extensive cell formatting, conditional styling, custom layouts, and interactive charts.
- **Multi-Worksheet Support:** Groups multiple tabs and data tables cleanly within a single workbook file.
- **Strict Data Types:** Retains exact data types (dates, text, numbers) without stripping formatting like leading zeros.

- **Cons:**

- **Larger File Sizes:** Embedded XML schema and styling metadata increase file overhead compared to flat text.
- **Software Dependency:** Requires Microsoft Excel, compatible spreadsheet apps, or specific backend libraries to process.
- **Poor Version Control:** Compressed file package makes line-by-line git diff tracking impossible.
- **Resource Heavy:** Reading massive datasets in XLSX format consumes significantly more memory and processing power.

3. Learn about Encoding

Encoding is the process of translating human-readable information—such as text, numbers, images, audio, or video—into a standardized digital format composed of binary 0s and 1s. This conversion allows computer systems to store, process, and transmit data reliably across different hardware and network protocols without corruption. Rather than hiding or securing data, encoding simply restructures it using publicly known rules so that receiving systems can easily decode it back to its original state.

Main Types of Encoding & Real-World Implementations

1. Character Encoding (Text to Binary) Translates characters and symbols into numeric byte sequences so software can display text correctly across platforms.

- **ASCII:** A basic 7-bit standard used in legacy command-line interfaces.
 - *Example:* The character A maps to decimal 65, stored as binary 01000001.
- **UTF-8:** The universal variable-length standard powering over 98% of the web.
 - *Example:* The letter A takes 1 byte (01000001), while the emoji 🤪 takes 4 bytes.

- **Real-World Use:** Web browsers use UTF-8 headers to load multilingual web pages and render English, Asian scripts, and emojis seamlessly on the same screen.

2. Binary-to-Text Encoding (Safe Transport) Converts raw binary data into safe ASCII text characters so it can pass through text-only transmission channels without breaking parser rules.

- **Base64:** Translates binary files into text strings made of 64 safe ASCII characters.
 - *Example:* The word Hello encodes to SGVsbG8=.
- **URL Encoding:** Replaces unsafe characters in links with % followed by hex codes.
 - *Example:* Searching for c++ tutorial converts to c%2B%2B%20tutorial.
- **Real-World Use:** Email protocols use Base64 to attach images or PDFs inside plain-text email streams, while APIs use it to send image files inside JSON payloads.

3. Media Encoding & Codecs (Audio/Video) Compresses raw sound waves or video frames into efficient streaming formats.

- *Example Formats:* MP3/AAC for audio; H.264 (AVC) and AV1 for video.
- **Real-World Use:** Platforms like YouTube encode high-resolution raw footage into streaming files optimized for different internet bandwidths.

4. Create an account on 'Hugging face'

Hugging face id url

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